Hardware – Algorithm Compatibility & Real-Time Performance Report

aBAJA SAEINDIA 2026 | Autonomous Vehicle System

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1. Purpose

This report satisfies the **Hardware-Software Compatibility Deliverable** of the aBAJA SAEINDIA 2026 submission guidelines. It demonstrates that the proposed autonomous algorithms — **Lane Keeping Assist (LKA)**, **Autonomous Emergency Braking (AEB)**, and **sensor fusion-based state estimation (EKF)** — are compatible with the selected embedded compute platform (**NVIDIA Jetson Orin Nano 8 GB**), meet the required real-time performance thresholds (Perception ≥ 10 Hz, Control ≥ 20 Hz), and comply with sensor and compute budget constraints. The platform is physically deployable within the aBAJA vehicle chassis and is commercially available in the Indian market.

2. Proposed Compute Platform

The target deployment platform is the **NVIDIA Jetson Orin Nano 8 GB Developer Kit**. This platform provides GPU-accelerated compute for the multi-sensor perception and control stack, is mountable within the aBAJA vehicle electronics bay, and is commercially available within budget. The platform runs Ubuntu Server 22.04 integrated with ROS 2 Humble middleware and is powered via a regulated 19 V vehicle-compatible supply system.

Component	Specification
Platform	NVIDIA Jetson Orin Nano 8 GB Developer Kit
CPU	6-core Arm Cortex-A78AE @ up to 1.5 GHz
GPU	1024-core NVIDIA Ampere architecture GPU
RAM	8 GB LPDDR5
Storage	microSD / NVMe SSD
OS	Ubuntu Server 22.04 (JetPack 6.x)
Middleware	ROS 2 Humble (DDS-based, low-latency)
Power consumption	7–15 W (compatible with regulated 19 V vehicle supply)
Form factor	~100 × 80 mm — mountable in vehicle electronics bay

3. Algorithm Computational Analysis

The autonomous stack executes within the compute constraints of the Jetson Orin Nano. Three primary algorithm pipelines run concurrently: Lane Keeping Assist (LKA), Autonomous Emergency Braking (AEB), and EKF-based sensor fusion for vehicle state estimation.

3.1 LKA — Per-Frame Operation Breakdown

Camera images are processed at 30 FPS. One control cycle budget at 30 Hz = 33.3 ms. The breakdown below estimates per-frame compute cost on the Cortex-A78AE CPU.

Pipeline Stage	Operations	Approx. Cost
Frame capture / buffer	Camera frame grab, DMA transfer	~0.5 ms
Grayscale + threshold	640×480 pixel ops (OpenCV)	~0.3 ms
BEV warp	cv2.warpPerspective, bilinear interpolation	~1.2 ms
Lane pixel extraction	Sliding window search / histogram centroid	~1.0 ms
Polynomial fit	np.polyfit, 1st-order, ~200 pts per lane	~0.5 ms
Lateral error compute	Lane centre deviation calculation	~0.1 ms
PID steering control	Proportional-Integral-Derivative scalar ops	~0.1 ms
ROS 2 topic publish	DDS message serialisation and publish	~0.5 ms
Total estimated	**~8× headroom @ 30 Hz**	**~4.2 ms**

At 30 Hz, cycle budget = 33.3 ms. Estimated 4.2 ms leaves ~8× headroom. Even with OS scheduling jitter and DDS latency, the 30 Hz target is comfortably achievable.

3.2 AEB — Per-Cycle Compute Breakdown

Pipeline Stage	Operations	Approx. Cost
Radar frame receive	UART / CAN read, buffer copy	~1.0 ms
Obstacle distance parse	Distance + relative velocity extraction	~0.2 ms
Fuzzy logic evaluation	Membership functions + rule inference	~0.3 ms
Brake command compute	Defuzzification → brake intensity scalar	~0.1 ms
ROS 2 publish	DDS serialise + publish to actuator node	~0.5 ms
Total estimated	**~24× headroom @ 20 Hz**	**~2.1 ms**

Radar processed at 20–30 Hz. At 20 Hz, cycle budget = 50 ms. Estimated 2.1 ms leaves ~24× headroom.

3.3 EKF Sensor Fusion — Per-Cycle Breakdown

Pipeline Stage	Operations	Approx. Cost
IMU data read	I2C/SPI read, buffer	~0.2 ms
Wheel encoder read	GPIO / encoder pulse count	~0.1 ms
EKF predict step	State transition + Jacobian compute	~0.5 ms
EKF update step	Kalman gain + state correction	~0.5 ms
State publish	ROS 2 nav_msgs/Odometry publish	~0.3 ms
Total estimated	**~6× headroom @ 100 Hz**	**~1.6 ms**

EKF targets 100 Hz. At 100 Hz, cycle budget = 10 ms. Estimated 1.6 ms leaves ~6× headroom.

3.4 Why PID + Fuzzy Logic Was Chosen Over Deep Learning (Current Phase)

Factor	PID + Fuzzy Logic (Selected)	Deep Learning (e.g. UFLD V2 + DNN Braking)
Inference speed	~200+ Hz (CPU only)	~30–60 Hz (GPU required)
GPU dependency	None — CPU-only execution	Requires TensorRT + GPU
Training data	None — parameter-based design	Requires domain fine-tuning dataset

Factor	PID + Fuzzy Logic (Selected)	Deep Learning (e.g. UFLD V2 + DNN Braking)
Development time	2–3 weeks	4–8 weeks (data + training)
Interpretability	Full — every stage inspectable	Black-box feature representations
Embedded deploy risk	Low — pure Python + OpenCV + ROS 2	Medium — TensorRT conversion needed
Robustness	Medium — tuning required	High — augmentation covers variation
Rulebook compliance	Full	Full — model inference compliant

Decision: PID + Fuzzy Logic selected for development timeline, embedded efficiency, and full interpretability. Deep learning (UFLD V2 / TensorRT) is the recommended upgrade path.

4. Budget Compliance — Itemised Cost Breakdown

All costs are indicative retail prices (INR) in the Indian market at time of submission. Actual procurement costs may vary. The total must be validated against the current aBAJA SAEINDIA 2026 rulebook budget cap.

Item	Specification	Cost (INR)
Compute platform	NVIDIA Jetson Orin Nano 8 GB Developer Kit	■26,000
Camera system	Automotive-grade camera, ≥30 fps, 640×480+	■2,000
Radar module	Short-range automotive radar, CAN/UART output	■60,000
IMU sensor	6-DOF IMU (accelerometer + gyroscope), I2C/SPI	■2,700
Wheel encoder system	Quadrature encoder, 2× (front wheels), GPIO interface	■1,000
Power regulation	19 V regulated DC supply, vehicle-compatible	■3,500
Wiring / connectors	USB, CAN, UART harness to sensors + vehicle	■2,000
Protective enclosure	ABS project box, IP44-rated	■2,500
Miscellaneous	Mounting hardware, cables, fuses	■5,000
TOTAL		**■104,700**

5. Deep Learning Upgrade Path (Future Work)

The Jetson Orin Nano 8 GB's Ampere GPU and 1024 CUDA cores make it significantly more capable for future neural network deployment. The recommended upgrade path is outlined below:

Step	Action
1	Collect 500–1,000 annotated frames from IPG CarMaker / CARLA simulation runs at correct exposure settings
2	Fine-tune UFLD V2 (Ultra-Fast Lane Detection) on collected dataset — pre-trained TuSimple/CULane weights available publicly
3	Export to ONNX → convert to TensorRT FP16 for Jetson Orin Nano GPU inference
4	Benchmark inference latency on Jetson Orin Nano (target: <33 ms per frame at 30 Hz)
5	Integrate deep learning lane output into ROS 2 pipeline; replace sliding-window node; PID controller unchanged
6	Upgrade AEB with DNN-based obstacle classifier (camera + radar fusion) for improved detection robustness
7	Integrate LiDAR for SLAM-based localisation and advanced path planning in endurance events
8	Deploy TensorRT-accelerated inference pipelines for perception; validate RMSE against classical baseline

6. Real-Time Performance Summary

Requirement	Threshold	This System	Event	Status
Perception rate (camera)	≥ 10 Hz	30 Hz (camera pipeline)	LKA / AEB	✔ PASS
LKA control rate	≥ 20 Hz	50 Hz (PID control loop)	LKA	✔ PASS
AEB radar processing	≥ 10 Hz	20–30 Hz (radar pipeline)	AEB	✔ PASS
EKF sensor fusion rate	≥ 50 Hz	100 Hz (IMU + encoder + radar)	Endurance	✔ PASS
Per-frame latency (LKA)	< 100 ms	~4.2 ms estimated	LKA	✔ PASS
Per-frame latency (AEB)	< 100 ms	~2.1 ms estimated	AEB	✔ PASS
Steering control latency	< 10 ms	< 10 ms (PID scalar ops)	LKA	✔ PASS
Average perception latency	< 35 ms	~4.2 ms (8× margin)	LKA / AEB	✔ PASS
Compute platform	Deployable + commercial	NVIDIA Jetson Orin Nano 8 GB — commercially available in India	All	✔ PASS

7. Event Coverage — AEB, LKA & Endurance

The table below summarises how the autonomous stack directly addresses each of the three graded events in aBAJA SAEINDIA 2026.

Event	Key Requirements Addressed	Subsystem(s)	Status
Lane Keeping Assist (LKA)	Camera-based BEV lane detection at 30 FPS; PID lateral control at 50 Hz; ~4.2 ms per-frame latency; ROS 2 topic p	Camera + OpenCV + PID + ROS 2 Humble	✔ COVERED
Autonomous Emergency Braking (AEB)	Short-range radar at 20–30 Hz; fuzzy-logic brake decision < 2.1 ms; defuzzified brake command published via ROS 2	Radar + Fuzzy Logic + ROS 2	✔ COVERED
Endurance (State Estimation)	EKF sensor fusion at 100 Hz fusing IMU (6-DOF) + wheel encoders; continuous vehicle state via ROS 2 nav_msgs/Odor	IMU + Wheel Encoder + EKF + ROS 2 Humble	✔ COVERED